

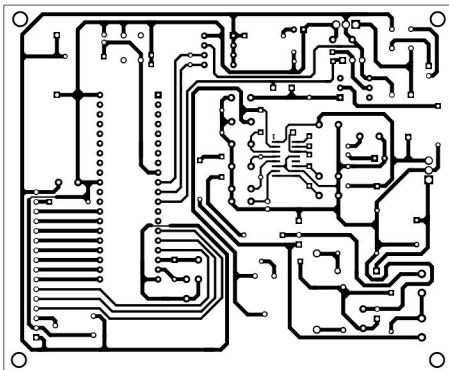


Fig. 4: An actual-size, single-side PCB for the energy meter

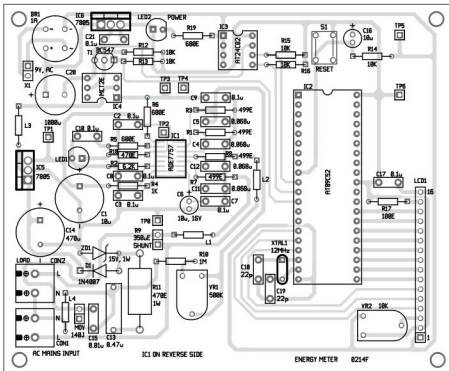


Fig. 5: Component layout for the PCB

Pins 15, 16 and 17 of IC2 are connected with the control pins RS, R/W and EN of LCD, respectively. Power-on reset is provided by the combination of resistor R14 and capacitor C16. Switch

S1 is used for manual reset. A 12MHz crystal along with two 22pF capacitors provide basic clock frequency to the microcontroller. Preset VR2 is connected with pin 3 of LCD for contrast

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com website.

control.

AT24C02. It is an FC-bus compatible 2-kilobit EEPROM organised as 256×8 bits that can retain data for more than ten years. To obviate the loss of latest setting in the case of power failure, the microcontroller can store all data of user in the EEPROM. The memory ensures that the microcontroller will read the last saved data from EEPROM when power resumes. Using SCL and SDA lines of EEPROM, the microcontroller can read/write the data from/to AT24C02 memory. SCL and SDA lines of IC3 are interfaced to pin 10 and 11 of microcontroller IC2, respectively.

Software

The source program for the microcontroller is written in C language and compiled using the 'Keil µVision4' compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. On reset/power-on, the microcontroller executes the main function. The interrupt INT0 and LCD are initialised first. Message 'Energy Meter' and reading are displayed in first and second lines on LCD module. Continuous loop in the code reads the data from EEPROM and displays in the second line of LCD module. Header file i2c.h is used for FC protocol communication and lcd.h is driver file for LCD.

Construction and testing

An actual-size, single-side PCB for the energy meter is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on the PCB to minimise assembly errors. IC1 is an SMD package so should be soldered on the solder side of the PCB. Use IC bases for other ICs. ●

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